

Q3		
(a)(i)	Fluid pressure increases with depth. The upward forces due to the fluid pressure acting on the lower surface of the material are larger than the downward forces due to the fluid pressure acting on the upper surface of the material, resulting in a net upward force called the upthrust.	B1 B1
(a)(ii)	Upthrust on cup = weight of water displaced $= V \rho_{\text{water}} g$ $= (6.8 \times 10^{-5}) \times (1000) \times (9.81) = 0.66708 \text{ N}$ Weight of cup = $V \rho_{\text{cup}} g$ $= (6.8 \times 10^{-5}) \times (2200) \times (9.81)$ $= 1.4676 \text{ N}$ Since the weight of the cup is larger than the upthrust acting on the cup, an <i>upwards</i> external force F is required to keep the cup stationary. Since the cup is held in equilibrium, the net force acting on the cup is zero. Hence, $F + U = W_{\text{cup}}$ $F = W_{\text{cup}} - U$ $= 1.4676 - 0.66708$ $= 0.800496$ $= 0.80 \text{ N}$	M1 M1 A0
(a)(iii)	Since the cup was already fully submerged, there is no difference in the volume and hence weight of fluid displaced, the upthrust stays constant. or The difference in pressure between the upper and lower surface remains the same and therefore the upthrust stays constant.	B1 B1
(b)	Pressure of compressed air = pressure at depth d $= (\text{atmospheric pressure}) + (\text{hydrostatic pressure})$ $= p_{\text{atm}} + \rho g d$ $= (1.0 \times 10^5) + (1000) \times (9.81) \times (0.30)$ $= 1.02943 \times 10^5 \text{ Pa}$ Applying $p V = \text{constant}$, since temperature is unchanged, $(1.0 \times 10^5) (5.50 \times 10^{-4}) = (1.02943 \times 10^5) V$ $V = 5.3429 \times 10^{-4} \text{ m}^3$ $= 5.3 \times 10^{-4} \text{ m}^3$	B1 M1 A1

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